



CALICUT UNIVERSITY – FOUR-YEAR UNDER GRADUATE PROGRAMME (CU-FYUGP)

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	PYTHON BASICS				
Type of Course	Vocational Minor (GROUP II: DATA ANALYSIS IN PHYSICS)				
Semester	I				
Academic Level	100 - 199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Basic computer knowledge				
Course Summary	This course introduces Python programming for data analysis in Physics with the aid of machine learning. As the first step, Python language is introduced with emphasis on Numpy and matplotlib modules, for future use in machine learning.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the significance of algorithm & flowchart in development of computer programs	U	F	Instructor-created exams
CO2	Understand and apply basic Python syntax	U, Ap	F, P	Instructor-created exams, Practical Assignment / Observation of Practical Skills
CO3	Understand and apply various conditional statements, as well as	U, Ap	F, P	Instructor-created exams, Practical Assignment /

	understand the modular nature of a program using functions in Python.			Observation of Practical Skills
CO4	Apply various modules for several tasks in Python	Ap	P	Instructor-created exams, Practical Assignment / Observation of Practical Skills/ Home Assignments
CO5	Understand in detail and apply the Numpy module in data analysis of physical data.	U, Ap	F, P	Instructor-created exams, Practical Assignment / Observation of Practical Skills
CO6	Understand and apply the matplotlib module for graphical representation of data in various pictorial formats.	U, Ap, C	F, P	Instructor-created exams, Practical Assignment / Observation of Practical Skills/ Home Assignments
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	Introduction to Python		12	15
	1	Use of algorithm and flowchart in computation.	2	
	2	Introduction to python, interactive and script mode, operators	2	
	3	Data types: numeric, string, list, tuple, set, dictionary (basics)	2	
	4	List operations, input() function, print() function, different formatted print statements, type() and eval() functions.	3	
	5	Files in Python & file operations: opening in different modes, read and write operations	3	

Chapter 2: p.31-33 (including Python’s IDLE Graphics window), Chapter 3,4, Chapter 5: p.95-108 (upto and excluding Command Line Arguments), Chapter 17: p.441-452 (including with statement), from <i>Core Python Programming</i> .					
II	Control statements, Functions and Modules			10	15
	6	Conditional & control statements: if, if...else, if...elif else statements,		2	
	7	while and for loops, range() function. Nested loops. break & continue statements.		3	
	8	Functions: built-in functions & user defined functions,		3	
	9	Modules and Packages, lambda expressions. Calendar Module, Math Module, time module, date module, zip()		2	
Chapter 6: p.117-139, Chapter 9: p.237-270, Chapter 20: p.515-526 of Book 1					
III	Numpy			15	25
	10	Numpy Arrays: creating arrays using array(), linspace, logspace, arrange(), zeros() and ones() functions.		2	
	11	Mathematical operations on arrays.		2	
	12	Indexing and slicing arrays, dimension of array		1	
	13	Attributes of arrays: ndim, shape, size, itemsize, dtype, nbytes		1	
	14	reshape() and flatten() methods for arrays		1	
	15	Multi-dimensional arrays using array(), zeros() and ones() functions		2	
	16	Indexing and slicing multi-dimensional arrays.		2	
	17	Numpy matrix: creation, access, mathematical operations.		2	
	18	Matrix operations (eigenvalues, dot, determinant, transpose, inverse), random numbers, shape(), reshape() functions.		2	
Chapter 6 of Book 2					
IV	Matplotlib module			8	15
	19	Plotting, labelling, scale commands in matplotlib		2	
	20	subplot, axes, figure, commands in matplotlib		2	
	21	Plotting pie chart, histogram, line graph, scatter plot and bar graphs.		2	
	22	grid(), axhline(), axvline() commands.		2	
	Chapter 14 of Book 2				

V	PRACTICALS		30
	Conduct any 5 experiments from the given list and 1 additional experiment, decided by the teacher-in-charge, related to the content of the course. The 6 th experiment may also be selected from the given list.		
	1	Developing Algorithms for Formatted Printing - Printing of triangle or inverted triangle (Pyramid form), Binomial coefficients in Pyramid form, fibonacci series.	
	2	Create and print a 3×3 matrix using nested loop.	
	3	Solution of simultaneous equations using Numpy.	
	4	Generate calendar using Calendar module.	
	5	Plot trigonometric functions - sin, cos, tan, x ² , exp(x).	
	6	Write a program for the ATM Pin verification process	
	7	Diagonalize a 3x3 matrix and verify that by evaluating the eigenvalues. Also evaluate the eigenvectors for the matrix.	
Relevant sections from Book 1 & Book 2			
Books and References:			
1. Core Python Programming 2 nd edition or higher, Dr. R. Nageswara Rao, Dreamtech press, 2020 (Book 1)			
2. Machine Learning in Data Science using Python, Dr. R. Nageswara Rao, Dreamtech press, 2022 (Book 2)			

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO 4	PS O5	PS O6	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7
CO 1	0	0	0	0	1	0	1	0	0	1	0	0	0
CO 2	0	0	0	0	1	0	1	0	0	1	0	0	0
CO 3	0	0	0	0	1	0	1	0	0	1	1	0	0
CO 4	0	0	0	0	1	0	1	0	0	2	1	0	0
CO 5	0	1	0	0	2	1	1	0	0	2	2	0	0
CO 6	0	1	0	0	2	1	1	0	0	2	1	0	0

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓			✓
CO 3	✓			✓
CO 4		✓		✓
CO 5		✓		✓
CO 6			✓	